

Comparative Analysis of Image Restoration and Segmentation Techniques for Noisy Images

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Problem Statement

Digital images are often affected by different types of noise during image acquisition, transmission, storage, or processing. Noise can reduce image quality and make it difficult to identify important objects and boundaries in an image.

Different types of noise require different restoration techniques. For example, Gaussian noise produces random intensity variations, whereas salt-and-pepper noise produces isolated black and white pixels. Therefore, selecting an appropriate filtering technique is important for recovering useful information from degraded images.

In this project, the effect of **Gaussian noise and salt-and-pepper noise** on grayscale images is studied. Different spatial filtering techniques, including **5 × 5 Mean filtering, 5 × 5 Median filtering, and Adaptive Median filtering**, are applied to restore the noisy images.

The restoration techniques are compared using **Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM)**. After restoration, image segmentation and boundary detection are performed using **Otsu thresholding with morphological processing and Canny edge detection**. The detected boundaries are compared with the ground-truth boundaries available in the BSDS500 dataset using **Intersection over Union (IoU) and Dice Score**.

The project therefore investigates how the type of noise influences the choice of restoration technique and how restoration affects subsequent image segmentation.

Objectives

The main objectives of this project are:

1. **To study the effect of different types of noise on grayscale images**, particularly Gaussian and salt-and-pepper noise.
2. **To implement different image restoration techniques**, including:
 - 5 × 5 Mean filter
 - 5 × 5 Median filter
 - Adaptive Median filter
3. **To compare the restoration techniques quantitatively** using:
 - Mean Squared Error (MSE)
 - Peak Signal-to-Noise Ratio (PSNR)
 - Structural Similarity Index (SSIM)
4. **To identify the most suitable restoration technique for each type of noise** based on the evaluation metrics.
5. **To perform image segmentation and boundary detection** on the restored images using Otsu thresholding, morphological processing, and Canny edge detection.
6. **To compare detected image boundaries with BSDS500 ground-truth boundaries** using IoU and Dice Score.

7. **To analyze the strengths and limitations of the implemented restoration and segmentation techniques.**

These objectives follow the assignment requirement to implement restoration and/or segmentation, test the methods, compare their results, and discuss limitations and possible improvements.

Dataset and Image Description

Dataset Used

The project uses the **Berkeley Segmentation Dataset 500 (BSDS500)**.

BSDS500 is a dataset containing natural images along with human-provided segmentation/boundary annotations. The dataset is suitable for this project because it provides both actual images and corresponding ground-truth information that can be used to evaluate segmentation and boundary detection.

The dataset is divided into:

- **Training set**
- **Validation set**
- **Test set**

For this project, the **test images** were used for the main evaluation.

Dataset used in our experiment: Berkeley Segmentation Dataset 500 (BSDS500).

Number of test images evaluated: 30

The assignment requires the report to specify the dataset source, number of images, image characteristics, preprocessing and limitations.

Image Characteristics

The BSDS500 dataset contains natural images with different objects, textures, backgrounds and boundaries.

For this project, the images were converted into **grayscale images** before noise addition and restoration.

A representative image used during the experiment contains a **polar bear in a snowy environment**. This image demonstrates the complete processing pipeline from the original image to the noisy, restored and segmented outputs.

Preprocessing

The following preprocessing steps were performed:

1. Images were loaded from the BSDS500 test dataset.
2. The images were converted to **grayscale**.
3. Artificial noise was added to simulate degraded images.
4. Two types of noise were considered:

- Gaussian noise
- Salt-and-pepper noise

5. The noisy images were then passed through different restoration filters.

The assignment specifically permits Gaussian and salt-and-pepper noise along with mean, median and adaptive filtering techniques

Dataset Limitation

Although BSDS500 provides useful ground-truth boundary information, its images contain **complex natural scenes with multiple objects, textures and boundaries**. Therefore, exact pixel-level matching between automatically detected Canny edges and manually annotated boundaries can be challenging.

This limitation is particularly relevant to the relatively low IoU and Dice values obtained in our segmentation experiment and will be discussed in detail in the **Results and Limitations** sections.

Methodology

The overall workflow of the project is:

Input Image → Grayscale Conversion → Noise Addition → Restoration → Evaluation → Segmentation → Boundary Detection → Ground-Truth Comparison

Stage 1: Input and Preprocessing

Images from the BSDS500 test set are loaded and converted into grayscale images.

Stage 2: Noise Addition

Two types of noise are introduced:

- Gaussian noise – produces random intensity variations.
- Salt-and-pepper noise – produces random black and white pixels.

Stage 3: Image Restoration

Three spatial filtering techniques are applied:

- 5×5 Mean filter
- 5×5 Median filter
- Adaptive Median filter

Stage 4: Restoration Evaluation

The restored images are compared with the original images using:

- MSE – lower value indicates lower error.
- PSNR – higher value indicates better reconstruction quality.
- SSIM – higher value indicates greater structural similarity.

Stage 5: Segmentation

The restored images are processed using:

- Otsu thresholding
- Morphological processing
- Canny edge detection

Stage 6: Evaluation

The Canny-detected boundaries are compared with the BSDS500 ground-truth boundaries using:

- IoU
- Dice Score

Implementation

The project was implemented in Python using Google Colab.

Libraries Used

- NumPy – numerical and image array operations
- OpenCV – image processing, filtering and Canny edge detection
- scikit-image – image quality evaluation and processing
- SciPy – supporting image-processing operations
- Matplotlib – displaying images and graphs
- Pandas – storing and comparing experimental results

DIP Techniques Used

Category	Technique
Noise	Gaussian noise
Noise	Salt-and-pepper noise
Restoration	5 × 5 Mean filter
Restoration	5 × 5 Median filter
Restoration	Adaptive Median filter
Segmentation	Otsu thresholding
Post-processing	Morphological processing
Boundary detection	Canny edge detection
Restoration evaluation	MSE, PSNR, SSIM
Segmentation evaluation	IoU, Dice

The experiment was performed on 30 BSDS500 test images, and average restoration metrics were calculated to compare the filtering techniques.

Results

Restoration Results

The best restoration method for each noise type was selected based on the restoration metrics.

Noise Type	Best Restoration Method	Average MSE ↓	Average PSNR ↑	Average SSIM ↑
Gaussian	Mean 5×5	281.6071	24.6632 dB	0.6285
Salt-and-Pepper	Adaptive Median	131.1859	28.8163 dB	0.8569

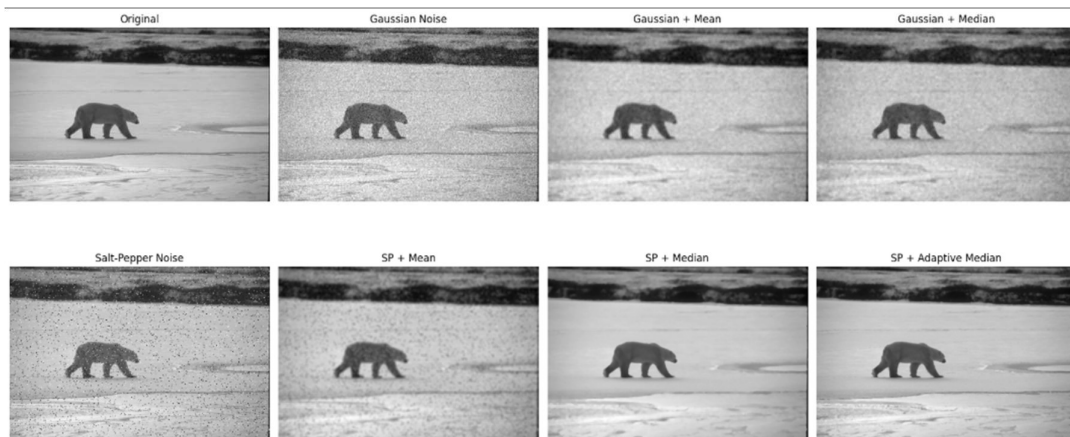
Analysis

For Gaussian noise, the 5×5 Mean filter was identified as the best-performing method.

For salt-and-pepper noise, the Adaptive Median filter produced the best result.

The salt-and-pepper restoration achieved a lower MSE and higher PSNR and SSIM compared with the Gaussian restoration result. This shows that the effectiveness of a restoration method depends on the type of noise present in the image.

Figure 1: Comparison of original, noisy, and restored images



Segmentation Results

The restored images were further processed using Otsu thresholding with morphological processing and Canny edge detection.

The detected Canny boundaries were compared with BSDS500 ground-truth boundaries.

Noise Type	Restoration Method	IoU ↑	Dice ↑
Gaussian	Mean 5×5	0.0317	0.0601
Salt-and-Pepper	Adaptive Median	0.0276	0.0525

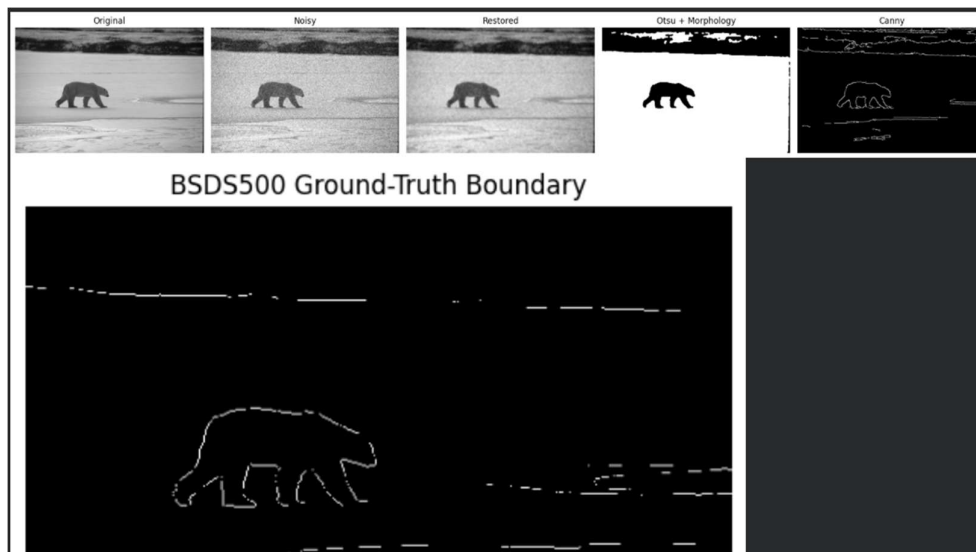
Analysis

The Gaussian + Mean restoration produced slightly higher IoU and Dice values than the Salt-and-Pepper + Adaptive Median case.

However, both IoU and Dice values were relatively low. This is mainly because the evaluation uses strict pixel-level comparison between Canny-detected boundaries and the BSDS500 ground-truth boundaries. Even small positional differences between the predicted and ground-truth boundaries can result in low overlap scores.

The visual output shows that important object boundaries, such as the polar bear in the sample image, can still be detected using Canny edge detection.

Figure 2: Segmentation and boundary detection results



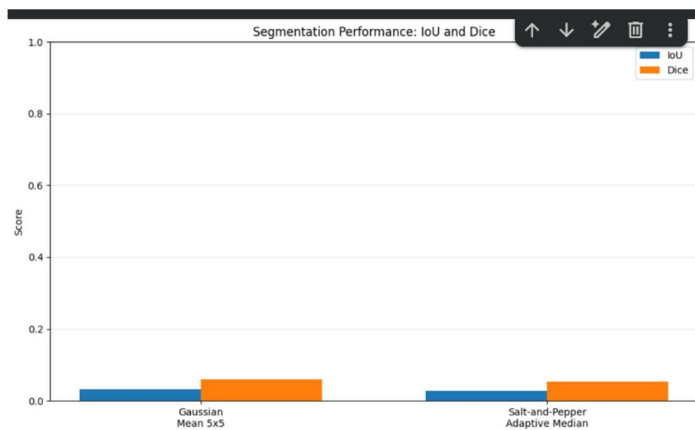
It contains:

Original → Noisy → Restored → Otsu + Morphology → Canny

and

BSDS500 Ground-Truth Boundary

Figure 3: Segmentation Performance using IoU and Dice



Limitations

The main limitations of the project are:

- The experiment was performed on only 30 test images.
- The project was limited to Gaussian and salt-and-pepper noise.
- Restoration performance depends on filter parameters such as window size.
- Mean filtering can cause blurring of edges and fine details.
- Segmentation of complex natural scenes is challenging.
- The IoU and Dice evaluation is sensitive to small positional differences between detected and ground-truth boundaries.
- Canny edge detection may detect additional edges that are not part of the main object boundary.

Future Scope

The project can be improved in the future by:

- Using a larger and more diverse image dataset.
- Testing additional noise types such as speckle and Poisson noise.
- Applying more advanced restoration techniques.
- Using deep learning-based image restoration and segmentation.
- Improving boundary evaluation using tolerance-based metrics.
- Developing a real-time image restoration and segmentation system.
- Deploying the system as a web or mobile application.

Conclusion

This project implemented and compared image restoration and segmentation techniques for noisy grayscale images using the BSDS500 dataset.

Gaussian noise and salt-and-pepper noise were introduced into the images, and Mean, Median, and Adaptive Median filters were applied for restoration. The restoration methods were evaluated using MSE, PSNR, and SSIM.

The experimental results showed that the 5×5 Mean filter performed best for Gaussian noise, while the Adaptive Median filter performed best for salt-and-pepper noise. The best Gaussian restoration achieved an average MSE of 281.6071, PSNR of 24.6632 dB, and SSIM of 0.6285. The best salt-and-pepper restoration achieved an average MSE of 131.1859, PSNR of 28.8163 dB, and SSIM of 0.8569.

Segmentation was performed using Otsu thresholding, morphological processing, and Canny edge detection. The resulting boundaries were evaluated using IoU and Dice Score. Although the quantitative segmentation scores were low, the visual results demonstrated the ability of the methods to identify important boundaries in natural images.

Overall, the project demonstrated that the choice of restoration technique should depend on the type of noise, and that both quantitative metrics and visual analysis are important when evaluating image processing systems.